DIGITAL DESIGN | FPGA | COMPUTER ARCHITECTURE

Iker Emiliano Garcia Barrera

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OVERVIEW

Electronics student (7th semester, UAA) focused on RTL design, computer architecture, and embedded systems. Experience designing and verifying hardware systems including RISC-V CPU components, IEEE-754 arithmetic units, and real-time embedded applications on STM32. Strong background in debugging at both hardware and software levels using waveform analysis and low-level tools. Interested in digital design and processor architecture roles.

TECHNICAL SKILLS

HDL & FPGA
Verilog (RTL Design), VHDL, Xilinx (ISE), Gowin EDA, Amaranth HDL

Embedded Systems
STM32, FreeRTOS, RFID, ARM Cortex-M4

Programming
C, Python, ARM Assembly, C++

Tools & Debug
Git, GTKWave, OpenOCD, GDB, KiCAD, COMSOL, Linux (CLI, Bash)

EDUCATION

Bachelor of Science in Electronics Engineering
Universidad Autónoma de Aguascalientes (UAA) • 7th Semester • Cambridge English B2 Certified

KEY PROJECTS

RISC-V CPU Architecture Model (GitHub)
Python/Amaranth HDL & Verilog

github.com/IKGB105/RISC-V-CPU-Designs

Designed and implemented a RISC-V architecture model with full instruction decode logic (opcode, registers, immediates). Implemented core datapath components including 32-bit ALU, dual-port RegisterFile (32×32), control unit, and instruction decoder. Debugged datapath and control inconsistencies using waveform analysis (GTKWave), isolating timing and logic errors across multiple test cases. Validated instruction execution through structured testbenches and simulation-driven verification.

IEEE-754 Floating-Point Calculator
VHDL / Xilinx Synthesis
Developed a single-precision (32-bit) IEEE-754 floating-point unit in VHDL as part of a team project. Collaborated on modular datapath design, integrating arithmetic units across team components. Implemented addition, subtraction, multiplication, and division. Handled normalization, rounding, and exception cases (NaN, overflow, underflow). Cross-validated hardware results with ARM Assembly implementations to ensure arithmetic correctness at low level.

RFID Access Control System (GitHub)
C / STM32F446 / ARM Cortex-M4

github.com/IKGB105/RFID-Access-Control-using-STM32F446RETx

Developed an embedded system for RFID-based access control with real-time processing. Collaborated in system integration and testing, ensuring reliable interaction between hardware and firmware components. Designed interrupt-driven architecture and state machines for reliable tag detection and system response. Built low-level drivers and handled peripheral communication on ARM Cortex-M4. Debugged hardware/software interaction using OpenOCD and GDB at register level.

Embedded Systems & Low-Level Programming
ARM Cortex-M4 Assembly & C/C++ / FreeRTOS / STM32
Developed ARM Cortex-M4 assembly and C/C++ programs focusing on direct hardware control and real-time systems. Implemented GPIO control, timers/PWM, UART communication, and interrupt service routines using register-level programming. Built FreeRTOS-based applications with multitasking, scheduling, and synchronization primitives. Analyzed and debugged concurrency issues and timing behavior in embedded environments.

LANGUAGES

Spanish — Native

English — B2 (Cambridge Certified)